

REPORT

Network Attack Log Analyzer

Objective

The objective of this project is to design and implement a **Network Attack Log Analyzer** that reads, processes, and analyzes network log files to detect suspicious and malicious activities. The system assists administrators and security analysts by generating alerts and reports from raw log data. It aims to enhance understanding of log analysis, cybersecurity fundamentals, and the application of data structures using Java and JavaFX.

Motivation

With the rapid growth of network-based systems, cyberattacks and unauthorized activities have become increasingly common. Network logs contain critical information that can help identify such threats; however, manual analysis is inefficient and error-prone. This project is motivated by the need for an automated and user-friendly tool that can analyze network logs, detect attack patterns, and present meaningful results visually. The system also

serves as an educational platform for learning JavaFX, object-oriented programming, and basic security monitoring techniques.

Concept

The concept of the **Network Attack Log Analyzer** is to create a lightweight desktop application that:

- Reads network log files from a selected directory
- Parses raw log data into structured objects
- Analyzes logs to detect malicious or suspicious behavior
- Generates alerts and summarized reports
- Displays results through a JavaFX graphical interface

The system integrates Java concepts such as MVC architecture, object-oriented design, data structures, and file handling to provide an effective log analysis solution.

Problem Statement

Modern network environments generate large volumes of log data, making it difficult to manually detect security threats.

Organizations face challenges such as identifying attack attempts, understanding abnormal traffic patterns, and responding to potential intrusions in a timely manner.

The problem addressed by this project is to design a simple yet effective system that:

- Automatically reads and processes network logs
 - Detects suspicious or malicious activities
 - Generates alerts and reports for analysis
 - Provides a graphical interface for better visualization
 - Uses efficient data structures for handling log data
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Design / Ways & Means

Introduction and Requirements

The system is developed in **Java** using **JavaFX** for the graphical user interface. It follows object-oriented principles and a modular design approach. The main requirements of the system include log file reading, attack detection, alert generation, report creation, and result visualization.

Data Structure Selection

The following data structures are used in the project:

- **ArrayList** → To store collections of log entries and alerts
- **HashMap** → For efficient mapping and tracking of log-related information

- **Custom Model Classes** → LogEntry, AlertEntry, ParsedLogEntry, and Report for structured data representation

These data structures ensure efficient storage, retrieval, and processing of log data.

Basic Implementation

The main components of the system include:

1. Log Reader

- Reads network log files from a user-selected directory
- Converts raw log data into structured objects

2. Log Analyzer

- Analyzes parsed logs
- Detects suspicious patterns and potential attacks
- Generates alerts and summary reports

3. JavaFX User Interface

- Allows users to select log directories
- Displays logs, alerts, and reports in table views
- Provides an interactive and user-friendly experience

4. Testing and Simulation Tools

- Used to verify functionality and integration of different modules

Performance Testing and Analysis

Performance evaluation of the system includes:

- **Log Reading:** Efficient file handling allows multiple log files to be processed smoothly
- **Log Analysis:** Analysis operations are linear in nature and suitable for moderate log sizes
- **Alert Generation:** Alerts are generated immediately upon detecting suspicious activity

Overall, the system performs efficiently for educational and small-scale analytical purposes.

Optimization and Advanced Features

Possible enhancements include:

- Integration with a database for persistent storage
- Real-time log monitoring
- Advanced attack detection algorithms
- Visualization dashboards and analytics
- Exporting reports in PDF or CSV format

Extensions and Creativity

Creative aspects of the project include:

- Modular service-based architecture
 - Separation of UI and business logic
 - Use of simulation and testing tools
 - Educational focus on cybersecurity and log analysis
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Analysis & Reporting

Lab Activity

The lab activity involved designing and implementing a network log analysis system using Java and appropriate data structures. The focus was on understanding log parsing, attack detection, JavaFX UI development, and system integration.

Background / Theory

Network log analysis is a fundamental concept in cybersecurity. Logs provide detailed records of network activity, which can be analyzed to detect intrusions, anomalies, and attack patterns. This project applies basic cybersecurity concepts such as monitoring, detection, and reporting in a simplified and educational manner.

Methodology

- Requirement analysis
- System design and architecture planning
- JavaFX UI development
- Implementation of log reader and analyzer
- Alert and report generation
- Testing and debugging
- Documentation and report preparation

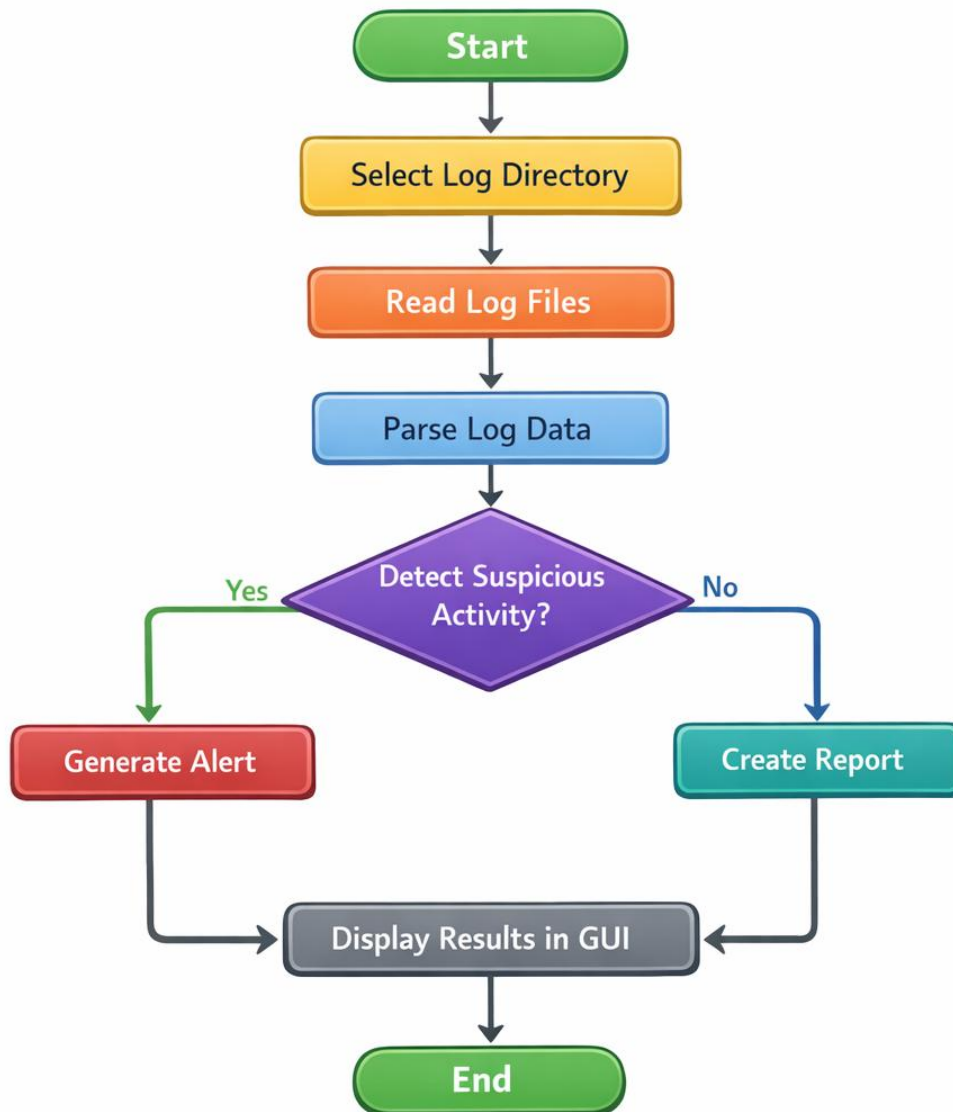
Data Collection

Network log files are collected from a selected directory and processed by the system. The logs are parsed and stored in structured data formats for analysis.

Block Diagram

The system flow follows these steps:

Network Attack Log Analyzer Flowchart



Analysis

- The system successfully reads and processes network logs
 - Suspicious activities are detected accurately
 - Alerts and reports are generated efficiently
 - The JavaFX interface improves usability and visualization
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Results

The project performs reliably under normal conditions. The analyzer effectively identifies suspicious patterns and generates meaningful alerts and reports. The system demonstrates practical application of Java, data structures, and basic cybersecurity principles.

Discussion on Results

The **Network Attack Log Analyzer** meets its objectives by providing an automated approach to network log analysis. It highlights the importance of log monitoring in cybersecurity and demonstrates strong understanding of object-oriented programming, data structures, and JavaFX development.